# Refined Urban Informal Settlements' Mapping at Agglomeration Scale With the Guidance of Background Knowledge From Easy-Accessed Crowdsourced Geospatial Data

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***Abstract*— Urban informal settlements (UISs) denote densely populated locales characterized by inadequate urban infrastructure standards, often exhibiting an amalgamation of rural and urban attributes, primarily situated within the confines of major cities or metropolitan regions. UIS mapping is a typical task that aims to identify pixels corresponding to UISs in remote sensing images. The extremely similar visual characteristics, sample uncertainty, and highly manual costs bring large-scale UIS mapping noteworthy challenges. In this article, we propose a sample uncertainty-aware semisupervised learning method guided by background knowledge from crowdsourced geospatial data (namely, SemiUIS) for UIS mapping and produces a 1-m resolution UIS map in the urban agglomeration in the middle reaches of the Yangtze River (UAMRYR), China. Our proposed SemiUIS method integrates semisupervised learning with the guidance of background knowledge from crowdsourcing data to reduce the uncertainty of the generated pseudosemantic segmentation samples to improve the quality of the constructed dataset by jointly optimizing the background (Not UIS areas) and the foreground (UIS areas). In contrast, traditional semisupervised learning methods only focus on explicitly optimizing the foreground (UIS areas) in the pseudosample generation process. Experiments were conducted in the 31 prefecture-level cities of UAMRYR. Visual interpretation of labeled remote sensing samples, street view images, and two novel knowledge validation indicators, global knowledge precision (GKP) and local knowledge precision (LKP), are proposed to verify the UIS mapping results. The proposed method reached an overall accuracy (OA) of 90.58%, mean intersection over union (mIoU) of 75.88%, accuracy (Acc) of 78.77%, Intersection over Union (IoU) of 62.97%, GKP of 90.33%, and LKP of 85.01%, respectively, with only 36 (5%) training samples. This work will be available at https://github.com/RunyuFan/UisYangtze.**

***Index Terms*— Crowdsourced geospatial data, deep learning (DL), remote sensing, semisupervised learning, urban informal settlement (UIS) mapping.**

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## I. Introduction

URBAN informal settlements (UISs) are high-density residential areas that accommodate the housing needs of a significant portion of the city's population. UISs are a specific focus in Sustainable Development Goal 11 (SDG 11) related to sustainable urban development. Currently, more than half of the world's population resides in urban areas, with a significant portion living in UIS [1]. Owing to deficiencies in urban planning, UIS often exhibits features such as low rise, deteriorating housing structures, traffic congestion, insufficient open spaces, and limited healthcare access, which collectively lead to subpar living conditions. Consequently, gaining comprehensive insights into the locations and geometrical boundaries of UIS becomes a crucial resource for advancing urban development and facilitating sustainable urban planning.

At the outset, the acquisition of UIS relied on on-site investigations conducted by governments. This method demands significant manual costs and resources, making it challenging to obtain timely and detailed information about UIS. Nowadays, the availability of high-spatial-resolution remote sensing images enables the capture of intricate physical and spatial structures, as well as the visual characteristics of ground objects [2], [3], [4]. In this case, there has been a growing interest in leveraging high-spatial-resolution images in conjunction with deep learning (DL)-based models for various urban computing applications, including urban remote sensing [5], [6], [7], [8], [9], [10], [11], urban functional zones mapping [12], [13], [14], land use and land cover mapping [15], [16], [17], and so on. In recent years, with the launch of various satellites, researchers have gained access to high-spatial-resolution remote sensing images, enabling the acquisition of information related to the location, spatial structures, and visual characteristics of UIS. In addition, diverse geographic spatial data obtained from the Internet-of-Things (IoT) devices have become a valuable data source for identifying UIS. The use of high-resolution remote sensing imagery, combined with artificial intelligence and machine learning methods, has attracted increasing attention for extracting UIS [3], [4], [18], [19], [20], [21], [22]. From a remote sensing perspective, morphological characteristics of UIS,

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such as small rooftop areas, low building heights, high building density, limited open spaces, narrow roads, and sparse vegetation, offer valuable clues for their identification [2], [23], [24], [25], [26]. High-resolution satellite remote sensing provides imagery with spatial resolutions of 1 m per pixel or higher, offering a solid foundation for differentiating small and heterogeneous land features (e.g., buildings and rooftops). Consequently, high-resolution satellite remote sensing imagery has gained more attention for the interpretation of remote sensing and geographic targets.

UIS mapping primarily utilizes traditional machine learning methods. These methods involve extracting specific semantic information from labeled training datasets to train efficient classifiers for identifying UIS [4], [27], [28]. Traditional machine learning methods are effective when expert-designed features are representative and comprehensive. However, manually designed features are often incomplete and costly to obtain and validate. Thus, traditional machine learning models struggle to meet accuracy requirements when the surrounding environment is complex.

In contrast, DL methods, which utilize powerful nonlinear deep neural networks, can automatically learn multilayered feature representations from input data without the need for predefined expert features. Models based on convolutional neural networks (CNNs) [18], [20], [29] are widely used for land parcel or grid-based extraction of UIS. These studies treat UIS mapping tasks as an image classification problem, where models use land parcels or grids as inputs to determine whether they contain UIS. However, image classification at the scale of land parcels or grids may not meet the requirements for fine-scale UIS mapping. Therefore, some researchers have focused on using semantic segmentation methods to extract UIS at the pixel level [2], [30], [31], [32]. Semantic segmentation assigns a predefined class label to every pixel in an image [33]. In this way, these methods can provide a more detailed representation than image classification methods that assign the same class label to an entire land parcel or image grid. Among various semantic segmentation methods, fully convolutional networks (FCNs), UNet, DeepLab models, and other semantic segmentation models [2], [30], [31], [32], [34] have achieved good results in UIS mapping compared to previous image classification methods, both in terms of accuracy and computational efficiency.

However, from the perspective of data annotation, whether the abovementioned image classification methods or semantic segmentation methods, they both map UIS in the supervised learning manner [2], [4], [27], [28], [30], [31], [32], [34], [35]. Notably, supervised learning-based methods obtained remarkable performances in UIS mapping by feature learning from a large number of well-labeled samples. However, in real-scene large-scale UIS mapping tasks, researchers usually face two major problems. First, obtaining pixel-level UIS labels is expensive and time-consuming. Second, due to the spatial heterogeneity of UIS, it is difficult to accurately label UIS and non-UIS through human-visual features of remote sensing images, especially in annotating semantic segmentation labels.

In view of the above problems, semisupervised learning methods and weakly supervised learning methods have obvious advantages in semantic segmentation tasks [36], [37], [38]. Semisupervised learning methods and weakly supervised learning allow the use of lower-cost and easy-access labels to reduce the cost of human annotation and reliance on a large number of annotations. The main advantages of these methods include reducing labeling costs and improving labeling efficiency. Semisupervised and weakly supervised learning can significantly reduce manual labeling costs, besides they are expected to improve the generalization ability of the model and make it more usable in different applications. The above advantages make semisupervised and weakly supervised learning methods more feasible to conduct research on large-scale area UIS mapping.

Based on the above ideas, this article proposes a sample uncertainty-aware semisupervised learning method guided by crowdsourced geospatial data samples for fine-grained agglomeration-scale UIS mapping, namely, SemiUIS. Specifically, a novel dual-path feature alignment transformer (DualFormer) model is proposed to address high spatial heterogeneity and regional mixing issues by optimizing the geometry and semantic information of UIS. Besides, focusing on the lack of labeled samples problem, we have proposed a semisupervised learning method to utilize large amounts of unlabeled data, and an automatic sample selection method DGUSL was proposed to select high-confidence pseudosemantic segmentation samples and generate a dataset from large amounts of unlabeled data automatically. To reduce the uncertainty of the generated pseudosemantic segmentation samples, we utilize background knowledge from crowdsourcing OpenStreetMap (OSM) databases together with AreaOfInterests (AOIs) from the Baidu map to improve the quality of the constructed dataset. The proposed semisupervised learning strategy together with the transformer was integrated into an end-to-end teacher–student manner to make our strategy easily transferable to any models and improve the classification accuracy.

Experiments were conducted in the urban agglomeration in the middle reaches of the Yangtze River (UAMRYR), China. Visual interpretation of labeled remote sensing samples, street view images, and two novel knowledge validation indicators, GKP and LKP, are proposed to verify the UIS mapping results. The proposed method reached an OA of 90.58%, an mIoU of 75.88%, an Acc of 78.77%, an IoU of 62.97%, a GKP of 90.33%, and an LKP of 85.01%, respectively, with only 5% training samples.

Compared with the recent works on UIS mapping, our work has the following three points of advantages and differences. First, our work maps UIS at the urban agglomerations scale and uses the visual interpretation of remote sensing images, sample points of street view images, and crowdsourced open geospatial data for multifaceted accuracy assessment. In contrast, other works [39], [40] map UIS in a single or a few cities or only use the visual interpretation of remote sensing images for validation. Second, our method uses a novel semisupervised learning strategy, which can greatly alleviate the problem of few labeled samples in large-area UIS mapping scenarios, making our method more suitable and efficient for urban agglomeration-scale UIS mapping than other fully supervised

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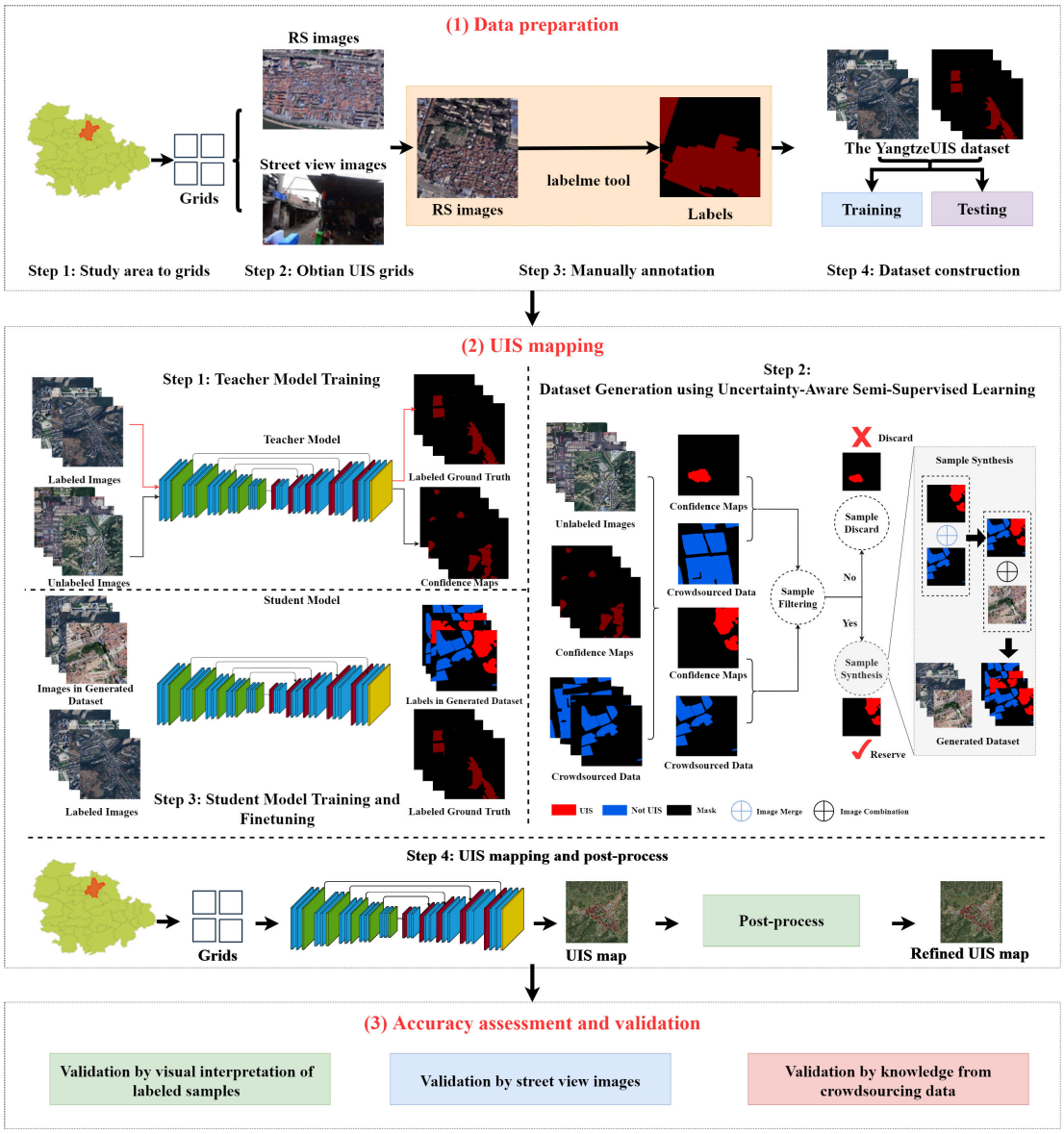



Fig. 1. Flowchart of the proposed UIS mapping method.

UIS mapping methods [4], [30], [41]. Third, our proposed automatic sample selection method integrates semisupervised learning with the guidance of background knowledge from crowdsourcing data to reduce the uncertainty of the generated pseudosemantic segmentation samples to improve the quality of the constructed dataset by jointly optimizing the background (Not UIS areas) and the foreground (UIS areas). In contrast, traditional semisupervised learning methods [36], [37], [38] only focus on explicitly optimizing the foreground (UIS areas) in the pseudosample generation process.

The main contributions of this article are listed as follows.

1) Focusing on the lack of labeled samples problem, we have proposed a semisupervised learning method to utilize large amounts of unlabeled data for UIS mapping at agglomeration scale.
2) An automatic sample selection method, namely, DGUSL, was adopted to select pseudosemantic segmentation samples and generate a dataset from large amounts of unlabeled data automatically.
3) Background knowledge from crowdsourcing data was used to reduce the uncertainty of the generated pseudosemantic segmentation samples to improve the quality of the constructed dataset.
4) Two novel knowledge validation indicators, GKP and LKP, are proposed to verify the UIS mapping results using only crowdsourcing data.

## II. Proposed Method for UIS Mapping

To tackle the lack of labeled samples problem, as well as to enhance the performance of the UIS mapping models, this article proposes a sample uncertainty-aware semisupervised learning method guided by crowdsourced weak samples for refined agglomeration-scale UIS mapping. The flowchart of the proposed SemiUIS method is shown in Fig. 1. It includes three steps: data preparation,

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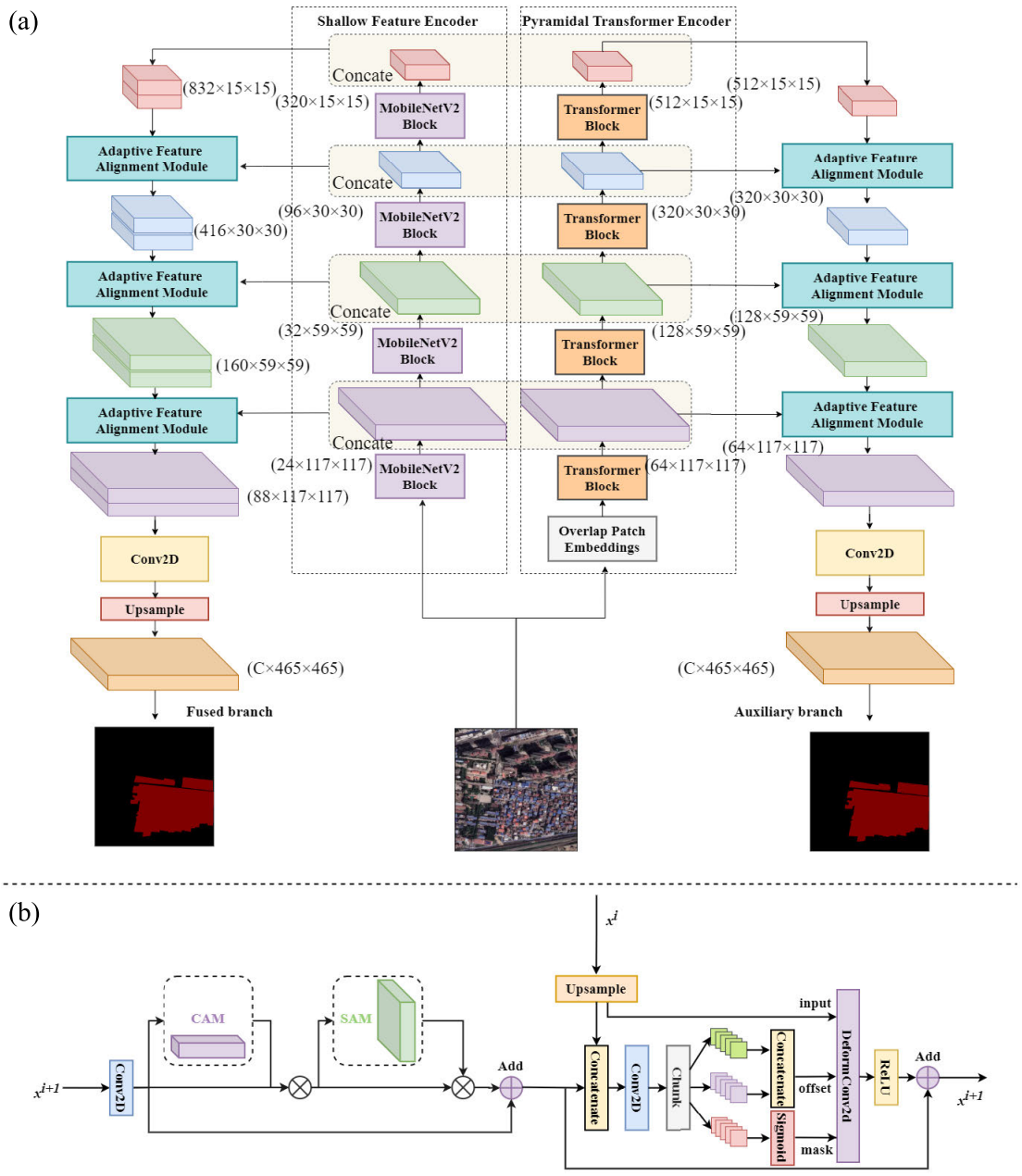



Fig. 2. Proposed DualFormer for UIS semantic segmentation. (a) DualFormer. (b) Adaptive feature alignment module.

UIS mapping, and accuracy assessment and validation. In the data preparation step, we first divided the whole study area into grids, then manually obtained UIS grid samples by referring to remote sensing and street view images, and manually labeled the UIS areas to construct the YangtzeUIS dataset. In the second step, we propose a novel DualFormer for UIS semantic segmentation, a semisupervised learning-based automatic sample selection method, namely, DGUSL, for selecting high-confidence pseudosemantic segmentation samples, utilizing background knowledge from crowdsourced geospatial data to generate dataset from large amounts of unlabeled data automatically, and then use the trained DualFormer model to map UIS and adopt postprocess to get the refined UIS map. In the third step, we perform accuracy assessment and validation.

### A. Architecture of DualFormer

This article introduces DualFormer, a novel UIS mapping method consisting of the shallow feature encoder (SFE), the pyramidal transformer encoder (PTE), and the adaptive feature alignment decoder (AFAD). Fig. 2 illustrates the overall network architecture and flowchart of DualFormer.

*1) Shallow Feature Encoder:* The proposed SFE uses lightweight and shallow MobileNetV2 [42] to extract spatial feature pyramids at different scales to capture multiscale low-level semantic context. MobileNetV2 is lightweight and shallow; thus, the feature pyramids from different layers of MobileNetV2 can well capture low-level features, such as color and shape contours, and can serve as a good complement to deep feature pyramids. The SFE module takes the remote

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sensing image as input, with the shape of $(3, H, W)$, and outputs feature pyramids of different scales, with sizes, respectively, $(24, H/4, W/4)$, $(32, H/8, W/8)$, $(96, H/16, W/16)$, and $(320, H/32, W/32)$.

In SFE, considering the efficiency and parameter costs, we use lightweight MibileNetV2 as the feature extractor and obtain feature pyramids to learn appropriate feature expressions in different scales. As a lightweight CNN, MibileNetV2 uses the idea of inverted residuals to improve storage efficiency. Besides, the MibileNetV2 model uses linear bottleneck and depthwise convolution to replace traditional convolution operations to ensure calculation accuracy while reducing parameters and can greatly increase the calculation speed.

*2) Pyramidal Transformer Encoder:* PTE uses a basic transformer to extract spatial feature pyramids at different scales to capture multiscale semantic context to make up for the scale variations. The PTE consists of an overlap patch embedding layer and four transformer blocks to hierarchically extract different scales of spatial features from the input remote sensing images. As shown in Fig. 2(a), in DualFormer, the PTE adopts the mix-transformer (MiT) [43] as backbone, takes an image of shape $(C_0, H, W)$ as input, and produces hierarchical multiscale features with the shape of $[C_1, (H/4), (W/4))$, $(C_2, (H/8), (W/8))$, $(C_3, (H/16), (W/16))$, and $(C_4, (H/32), (W/32)]$ from four stacked transformer blocks in the MiT backbone, respectively. In the transformer block, the input $x \in \mathbb{R}^{C_0 \times H \times W}$ is first fed into an overlap patch embeddings layer, which is defined as

$$x = \text{LayerNorm}(\text{Transpose}(\text{Flatten} \cdot (\text{Conv}_{C_i \times C_{i+1}}(x)))), i \in \{0, \ldots, 3\} \quad (1)$$

where $\text{Conv}_{C_i \times C_{i+1}}$ means a 2-D convolutional layer with inchannels of $C_i$ and outchannels of $C_{i+1}$. The kernel size, stride, and padding of $\text{Conv}_{C_i \times C_{i+1}}$ are set to $7 \times 7$, 4, and 3 when $i$ is 0 and are set to $3 \times 3$, 2, and 1 when i is not 0.

The overlap patch embedding layer then uses a 2-D convolutional layer to split the image $x$ into several 3-D patches. These 3-D patches then are flattened and transposed to 2-D patches, denoted as $x_p^i \in \mathbb{R}^{N \times (P_i^2 \times C_i)}$, where $N = ((H \times W)/P_i^2)$, $P_i$ is set to $(H/(4 \times 2^i))$, and $i \in \{0, \ldots, 3\}$. Then, a LayerNorm is added to $x_p^i$.

After that, $M$ identical self-attention (SA) layers [43] with residual connections are added. SA is formulated as

$$\text{SA}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_{\text{head}}}}\right)V. \quad (2)$$

The tensor output above is then fed to a feedforward network to provide positional information for transformers. The feedforward network is defined as

$$x^i = \text{MLP}\left(\text{GELU}\left(\text{DWConv}\left(\text{MLP}\left(x_p^i\right)\right)\right)\right) + x_p^i \quad (3)$$

where $x_p^i$ is the feature from the $i$th SA. DWConv denotes the depthwise convolutions with $3 \times 3$ convolution kernels. $x^i$ is then passed into LayerNorm, Reshape, and Permute operators continuously to obtain 3-D feature maps from 2-D patches. Eventually, the feature maps $x^i$ from the stage $i$ transformer blocks together form the pyramidal feature maps. In this study, $H$ and $W$ are set to 465. $C_0, C_1, C_2, C_3$, and $C_4$ are set to 3, 64, 128, 320, and 512. $R_0, R_1, R_2$, and $R_3$ are set to 64, 16, 4, and 1.

*3) Adaptive Feature Alignment Decoder:* After the SFE and PTE, we have obtained multiscale pyramidal feature maps $x^i$ of every encoder that contains valuable information about UIS. For each encoder, given the pyramidal feature maps $x^i$ from the above PTE or SFE as input, the AFAD utilizes AFAM to select and fuse the pyramidal feature maps across stages.

The architectures of the AFAM decoder are illustrated in Fig. 2(b). The AFAM decoder hallucinates higher resolution features by upsampling feature maps from higher pyramid levels. Then, these feature maps are laterally connected with feature maps from the bottom–up pathway to produce enhanced feature maps. In the AFAM decoder, the input feature map $x^{i+1}$ is first passed into a $1 \times 1$ conv2d layer followed by cascaded CAM and SAM modules [44] with shortcut connection to learn the importance of each feature map channels. The detailed procedure can be formulated as follows:

$$x^{i+1} = \text{CAM}(x^{i+1}) \otimes x^{i+1} \quad (4)$$

$$x^{i+1} = \text{SAM}(x^{i+1}) \otimes x^{i+1} \quad (5)$$

$$\text{CAM}(x^{i+1}) = \phi(\text{MLP}(\text{AvgPool}(x^{i+1})) + \text{MLP}(\text{MaxPool}(x^{i+1}))) \quad (6)$$

$$\text{SAM}(x^{i+1}) = \phi(\text{Conv}_{7\times7}([\text{AvgPool}(x^{i+1}) \text{MaxPool}(x^{i+1})])) \quad (7)$$

where $\phi$ denotes the sigmoid function, AvgPool denotes the average-pooling operation, MaxPool denotes the max-pooling operation, and $\text{Conv}_{7\times7}$ represents a convolution operation with the kernel size of $7 \times 7$.

Then, the feature maps $x^{i+1}$ above and the feature maps $x^i$ from the pyramidal feature from the encoder are aligned and fused. Due to the different scales of $x^i$ and $x^{i+1}$, there is a spatial misalignment, which would harm the prediction of UIS when fusing the above feature maps. The deformable convolution is studied, which can be used as a feature alignment function, owing to its capability of using additional offsets and learning the offsets to augment the spatial sampling locations [45], [46]. Specifically, the offset value can be calculated by the shifted distances between each point in $x^i$ and its corresponding point in $x^{i+1}$. The detailed procedure can be mathematically formulated as

$$x^i = \text{Upsample}(x^i) \quad (8)$$

$$x_1^{\text{fuse}}, x_2^{\text{fuse}}, x_3^{\text{fuse}} = \text{Chunk}(\text{Conv2D} \cdot (\text{Concatenate}(x^i x^{i+1}))) \quad (9)$$

$$\text{offset} = \text{Concatenate}\left(x_1^{\text{fuse}}, x_2^{\text{fuse}}\right) \quad (10)$$

$$\text{mask} = \text{Sigmoid}\left(x_3^{\text{fuse}}\right) \quad (11)$$

$$x^{i+1} = \text{ReLU}(\text{DeformConv2d} \cdot (x^i, \text{offset}, \text{mask})) + x^{i+1}. \quad (12)$$

In the DualFormer, the pyramidal feature map from SFE $x_{\text{SFE}}$ is concatenated with the pyramidal feature map from PTE $x_{\text{PTE}}$ to a fused pyramidal feature map $x_{\text{Fuse}}$. $x_{\text{PTE}}$ and $x_{\text{Fuse}}$ are fed into two individual AFAM decoders and obtained two feature maps with the same shape of $(C1 \times 117 \times 117)$.

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**Algorithm 1** DGUSL

**Input:** The confidence threshold: $\tau$, Unlabelled data: $U = \{u_1, u_2, \ldots, u_I\}$, Crowdsourced data: $C = \{c_1, c_2, \ldots, c_I\}$, The teacher model: $T$ **Output:** $NewDataSet\ D = \{d_1, d_2, \ldots, d_K\}$

**function** DGUSL($\tau$, $U$, $T$)
  $D \leftarrow \varnothing$
  **for** $i = 1$ to $I$ **do**
    Input $u_i$ into the model $T$ and calculate the corresponding confidence map, and denote it as $confidence^i_{teacher}$
    **for** confidence value in $confidence^i_{teacher}$ **do**
      **if** confidence value $\geq \tau$ **then**
        Update the corresponding confidence value to 1.
      **else**
        Update the corresponding confidence value to 0.
      **end if**
    **end for**
    *Sample Filtering:* Calculate the area of 1 in $confidence_{teacher}$, denote as $Area_i$.
    **if** $Area_i \geq 3000\ \mathrm{m}^2$ **then**
      Calculate the pixel number $N$ that overlaps the position of 1 in $confidence_{teacher}$ and the position of 1 in crowdsourced data $c_i$. And the total number of pixels of $confidence^i_{teacher}$ is denoted as $M$.
      **if** $N \div M < 0.5$ **then**
        *Sample Synthesis:* Generate a new sample $d_i$, update the value of the pixel in $d_i$ to 0 where the pixel of the same position in $confidence^i_{teacher}$ is 1. Update the pixel value in $d_i$ to 1 where the pixel of the same position in $c_i$ is 1, and update the value of the other pixels in $d_i$ to 2.
        Add $d_i$ to $D$
      **else**
        *Sample Discard*
      **end if**
    **else**
      *Sample Discard*
    **end if**
    continue
  **end for**
  **return** $D$
**end function**

The above two feature maps are then passed into two $1 \times 1$ convolutional layers with outchannels set to Cls, and a bilinear interpolation is used to rescale the outputs to the same width and length as the input image, that is, (Cls $\times$ 465 $\times$ 465). In this study, $C1$ is set to 128. Cls is set to 2 for the supervised learning stage (teacher model training in Section II-B1) and 3 for the semisupervised learning stage (student model training in Section II-B3). The output from $x_{\mathrm{PTE}}$ is denoted as $y_{\mathrm{PTE}}$, and the output from $x_{\mathrm{Fuse}}$ is denoted as $y_{\mathrm{Fuse}}$.

*4) Hybrid Loss Function:* In this study, we defined a hybrid loss function, noted as HL($\cdot$), by the combination of the focal loss [47] and the auxiliary loss. The hybrid loss function is defined as follows:

$$\mathrm{HL}(\cdot) = \alpha * \mathrm{FL}(p_t(y_{\mathrm{PTE}}, y)) + (1 - \alpha) * \mathrm{FL}(p_t(y_{\mathrm{Fuse}}, y)) \tag{13}$$

$$\mathrm{FL}(p_t) = -(1 - p_t)^{\gamma} \log(p_t) \tag{14}$$

$$p_t(p, y) = \begin{cases} p, & \text{if } y = 1 \\ 1 - p, & \text{otherwise} \end{cases} \tag{15}$$

where FL($\cdot$) represents the focal loss and y represents the label. $\gamma > 0$, and when $\gamma = 1$, focal loss works like a cross-entropy loss function. $\alpha$ generally ranges from [0, 1]. It can be set by the tradeoff between the output of the auxiliary branch $y_{\mathrm{PTE}}$ and the output of the fused branch $y_{\mathrm{Fuse}}$. In this study, $\alpha$ is set to 0.3, and $\gamma$ is set to 2.

### B. Uncertainty-Aware Semisupervised Learning Strategy

In the proposed algorithm, to label the unlabeled samples more efficiently and precisely, a semisupervised learning strategy was adopted, as shown in Fig. 1. First, a teacher DualFormer model was used to train the labeled training samples initially to obtain preliminary sample discrimination. Then, the teacher DualFormer model was applied to select samples and generate a new dataset using the proposed DGUSL strategy. Finally, the newly generated dataset together with the labeled dataset was used to train a student DualFormer model to get the final result. The semisupervised strategy is described as follows.

*1) Teacher Model Training:* In the proposed uncertainty-aware semisupervised learning strategy, the first step is to train the teacher model (DualFormer). A powerful teacher model can significantly reduce the error propagation

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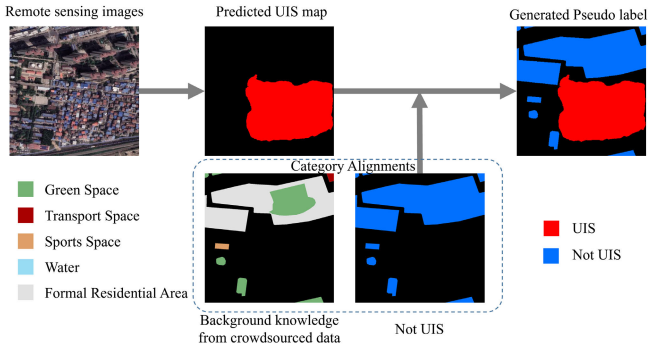



Fig. 3. Pseudolabel generation guided by background knowledge from crowdsourced geospatial data.

problem in the semisupervised sample selection process and directly affect the final performance. In this article, two strategies are used in the proposed DualFormer to guarantee a better teacher model. The first one is the combination and fusion of shallow features from the lightweight CNN-based MobileNetV2 model and pyramidal deep features from the transformer model. The second strategy is to use the AFAD to enhance the alignment and integration of features from different branches and different resolutions. The pyramidal features extracted and transformed by CNN models and transformer models are adaptively aligned to obtain feature maps with more geometric and semantic information. Through the above two strategies, a powerful teacher model is obtained for the semisupervised learning process.

*2) Dataset Generation Using Uncertainty-Aware Semisupervised Learning:* After the teacher model training step, we introduce the DGUSL to pseudolabel a large amount of unlabeled data and generate an effective and large-scale UIS training sample dataset. The process of DGUSL is shown in Algorithm 1. Here, the confidence threshold $\tau$, the unlabeled data $U = \{u_1, u_2, \ldots, u_I\}$, the crowdsourced data $C = \{c_1, c_2, \ldots, c_I\}$, and the teacher model $T$ are the inputs. The purpose of the DGUSL is to synthesize a generated dataset: New DataSet $D = \{d_1, d_2, \ldots, d_K\}$. In this step, first, the unlabeled sample is input into the DualFormer teacher model $T$ and calculate the corresponding confidence map, which is denoted as $\text{confidence}^i_{\text{teacher}}$. To reduce the error of the sample selection with DualFormer teacher model $T$, pixels in the confidence map are with a higher confidence value than the confidence threshold: $\tau$ is selected and updated to value 1, and the rest of the pixels are updated to value 0 (lines 4–11 in Algorithm 1). Then, we calculate the area of 1 in $\text{confidence}_{\text{teacher}}$, which is denoted as $\text{Area}_i$, and only reserve whose area is larger than 3000 $\text{m}^2$. Next, we calculate the pixel number $N$ that overlaps the position of 1 in $\text{confidence}_{\text{teacher}}$ and the position of 1 in crowdsourced data $c_i$. The total number of pixels of $\text{confidence}^i_{\text{teacher}}$ is denoted as $M$. We reserve those samples whose $N \div M < 0.5$ and generate a new sample $d_i$ by updating the value of the pixel in $d_i$ to 0 where the pixel of the same position in $\text{confidence}^i_{\text{teacher}}$ is 1, updating the pixel value in $d_i$ to 1 where the pixel of the same position in $c_i$ is 1, and updating the value of the other pixels in $d_i$ to 2 (lines 12–26 in Algorithm 1).

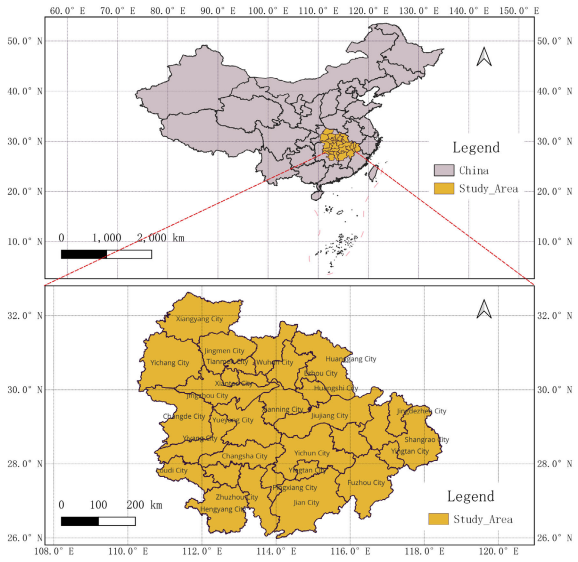



Fig. 4. UAMRYR.

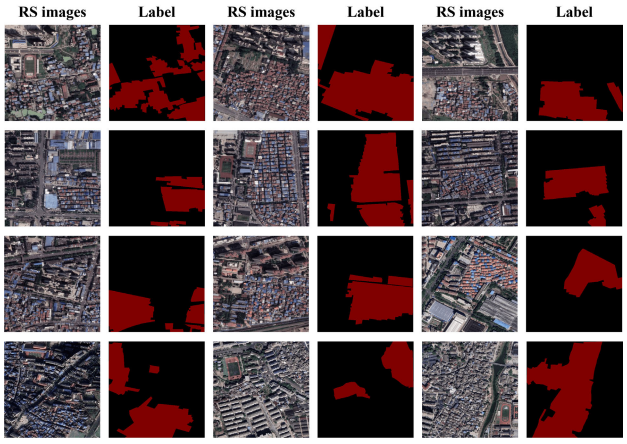



Fig. 5. Samples of labeled images in the YangtzeUIS dataset.

The general process of pseudosample generation can also be referred to in Fig. 3. In summary, we mapped the crowdsourced geospatial data to not UIS area. Then, we calculate the proportion of the samples that the high-confidence samples generated by the teacher model overlap with the formal settlements and delete the samples with overlap greater than 50%. Finally, the crowdsourced geospatial data are overlaid on the samples generated by the teacher model to generate new pseudosamples.

*3) Student Model Training:* The newly generated training dataset is then merged with the labeled UIS dataset to obtain a new dataset. The pixel values in the new dataset are 0, 1, and 2, which means UIS, Not UIS, and Mask. The student DualFormer model is finally trained with this newly generated training dataset. In the training process, label value 2 is ignored in our hybrid loss function, and in the testing process, the values in the channels that correspond to Mask are ignored; only the channels that correspond to UIS and Not UIS are calculated. Finally, the softmax features are used to calculate

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TABLE I
RESULTS FOR DIFFERENT MODELS AND THE PROPOSED METHOD

| Manner | Model Name | OA (%) | mIoU (%) | Acc of UIS (%) | IoU of UIS (%) | Parameters Amount |
|---|---|---|---|---|---|---|
| Supervised | FCN [48] | 88.92 | 71.90 | 71.63 | 56.78 | 32.95$M$ |
| Supervised | FPN [49] | 83.26 | 61.87 | 61.81 | 42.88 | 47.70$M$ |
| Supervised | FaPN [46] | 86.99 | 68.27 | 68.35 | 51.65 | 26.72$M$ |
| Supervised | UperNet [50] | 82.83 | 60.31 | 56.36 | 40.02 | 28.77$M$ |
| Supervised | SFNet [51] | 82.97 | 61.42 | 61.38 | 42.29 | 28.83$M$ |
| Supervised | DeeplabV3 [52] | 89.44 | 73.50 | 76.32 | 59.50 | 39.60$M$ |
| Supervised | Segformer [43] | 89.85 | 74.17 | 75.99 | 60.34 | 13.68$M$ |
| Semi-supervised | Fixmatch [53] | 89.33 | 72.74 | 72.32 | 57.99 | 40.50$M$ |
| Semi-supervised | Unimatch [54] | 89.95 | 73.94 | 73.00 | 59.68 | 40.50$M$ |
| Supervised | Our DualFormer | 90.09 | 74.84 | 77.70 | 61.45 | 18.63$M$ |
| Semi-supervised | Our SemiUIS | **90.58** | **75.88** | **78.77** | **62.97** | 18.63$M$ |

TABLE II
EXPERIMENTS OF DIFFERENT CONFIDENCE THRESHOLDS IN DGUSL

| Threshold | OA (%) | mIoU (%) | Acc of UIS (%) | IoU of UIS (%) |
|---|---|---|---|---|
| 0.5 | **90.58** | **75.88** | **78.77** | **62.97** |
| 0.6 | 90.57 | 75.57 | 76.65 | 62.30 |
| 0.7 | 90.39 | 75.08 | 75.59 | 61.52 |
| 0.8 | 90.13 | 73.49 | 68.31 | 58.45 |

the final categories, which was the output vector of the softmax layer (denoted as output $= \{a_1, a_2, \ldots, a_k\}$, and $k$ is the number of categories). Here,

$$a_i = \frac{e^{z_i}}{\sum_{j=0}^{k} e^{z_j}}, \quad i \in [0, k-1] \tag{16}$$

where $z$ is the input vector of the softmax layer and $k$ is set to 2, which represents UIS or Not UIS. The category corresponding to the largest value in output is the final predicted category.

### C. UIS Mapping and Postprocess

Based on the UIS semantic segmentation results of the proposed Dualformer in the divided grids, we further calculate the area of each UIS and delete the predicted UIS areas that are less than 5000 m$^2$. Finally, we use the "EliminatePolygonPart" tool in ArcGIS to fill the holes in the UIS polygons to obtain the final UIS mapping results. In this way, we produced the first very-high-spatial resolution (1.19 m/pixel) UIS map of urban agglomerations in the middle reaches of the Yangtze River.

### D. Two Novel UIS Mapping Validation Indicators

In order to assess and validate the accuracy of UIS mapping results, we utilize knowledge from third-party crowdsourcing data and propose a validation method by knowledge from crowdsourcing data. We proposed two validation indicators, global knowledge precision (GKP) and local knowledge precision (LKP), to verify the UIS mapping results. GKP and LKP are defined as follows:

$$\mathrm{GKP} = 1 - \frac{\sum_i^N \mathrm{UIS}_i \bigcap \mathrm{NotUIS}}{\sum_i^N \mathrm{UIS}_i}, \quad i \in [1, N] \tag{17}$$

$$\mathrm{LKP} = 1 - \frac{1}{N} \times \sum_i^N \frac{\mathrm{UIS}_i \bigcap \mathrm{NotUIS}}{\mathrm{UIS}_i}, \quad i \in [1, N]. \tag{18}$$

$\mathrm{UIS}_i$ is the $i$th predicted UIS regions and $i$ ranges from 1 to $N$. $N$ represents the total number of predicted UIS regions. NotUIS represents the non-UIS regions in merged polygons from AOI and OSM. $\bigcap$ represents the intersection of two regions. The larger the GKP and LKP, the higher the UIS mapping accuracy.

## III. EXPERIMENT AND ANALYSIS

### A. Study Area

The UAMRYR (as shown in Fig. 4) is a very large urban agglomeration, which was mainly formed by the urban agglomeration around Wuhan City, the urban agglomeration around Changsha City, and the urban agglomeration around Poyang Lake with an area of 326 100 km$^2$. It is an important part of the Yangtze River Economic Belt and a key area for implementing the strategy of promoting the rise of the central region, deepening reform and opening up in an all-round way, and promoting new urbanization. During the past decades, it has retained many UIS areas where old, tiny, and dilapidated residents gather densely and severely restricted sustainable development. Therefore, in this study, the UAMRYR is treated as a research area for UIS mapping research.

### B. YangtzeUIS Dataset

To better map the location of UIS in the UAMRYR, we used submeter high-resolution remote sensing images from Google Earth and manually constructed a meter-resolution (1.19 m) UIS semantic segmentation dataset (namely, YangtzeUIS dataset). As shown in Fig. 5, we obtained very-high-resolution (VHR) remote sensing images from Google Earth with a spatial resolution of 1.19 m. The VHR remote sensing images are cropped to the shape of 465 × 465, which indicates 550 m in the real landscape. Then, we obtained the semantic segmentation dataset of UIS through pixel-level labeling using

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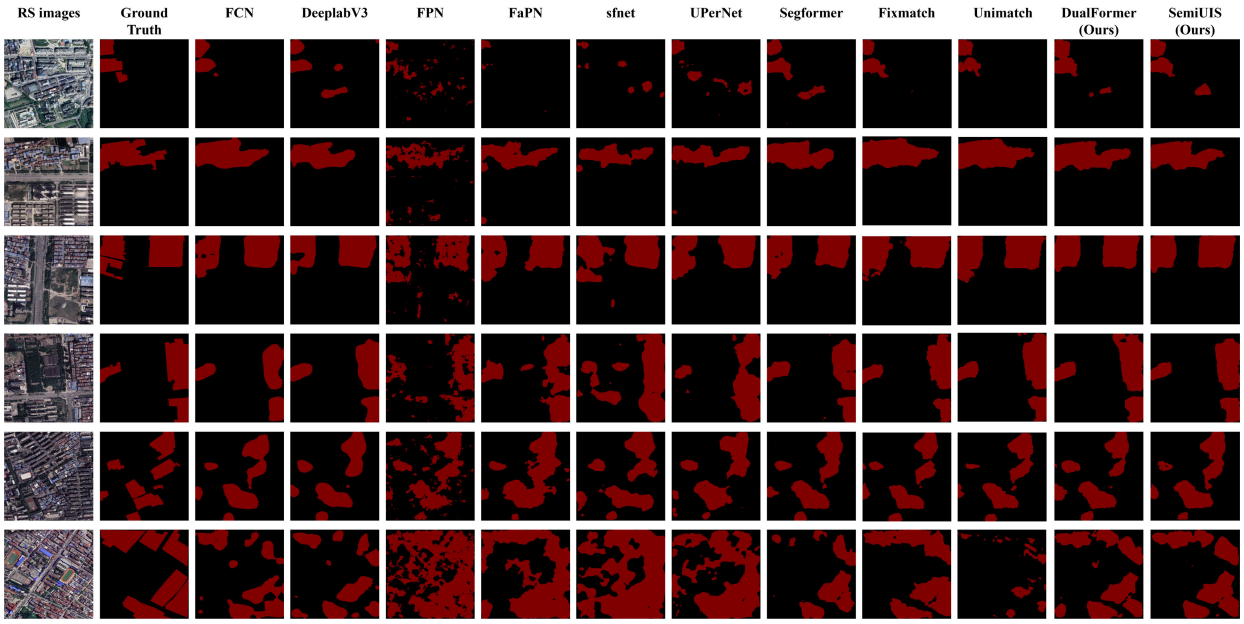



Fig. 6. Illustrations of some UIS semantic segmentation results.

TABLE III

ABLATION STUDY OF DIFFERENT COMPONENTS OF THE PROPOSED METHOD. THE BLUE SUBSCRIPT REPRESENTS THE DECREASED VALUE. THE RED SUBSCRIPT REPRESENTS THE GAINED VALUE

| Methods | SFE | PTE | AFAM | DGUSL | OA (%) | mIoU (%) | Acc of UIS (%) | IoU of UIS (%) |
|---|---|---|---|---|---|---|---|---|
| model1 (SFE + FPN Decoder) | √ | | | | 84.38 | 63.10 | 60.43 | 44.02 |
| model2 (PTE + FPN Decoder) | | √ | | | $83.99_{(-0.39)}$ | $63.21_{(+0.11)}$ | $63.93_{(+3.5)}$ | $44.81_{(+0.79)}$ |
| model3 (SFE + PTE + FPN Decoder) | √ | √ | | | $84.83_{(+0.84)}$ | $64.05_{(+0.84)}$ | $62.21_{(-1.72)}$ | $45.46_{(+0.65)}$ |
| DualFormer (Ours) | √ | √ | √ | | $90.09_{(+5.26)}$ | $74.84_{(+10.79)}$ | $77.70_{(+15.49)}$ | $61.45_{(+15.99)}$ |
| SemiUIS (Ours) | √ | √ | √ | √ | **90.58**$_{(+0.49)}$ | **75.88**$_{(+1.04)}$ | **78.77** $_{(+1.07)}$ | **62.97**$_{(+1.52)}$ |

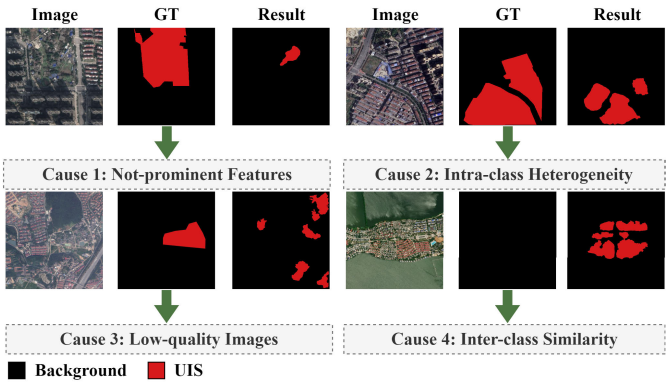



Fig. 7. Illustration of four representative causes that lead to UIS misidentification.

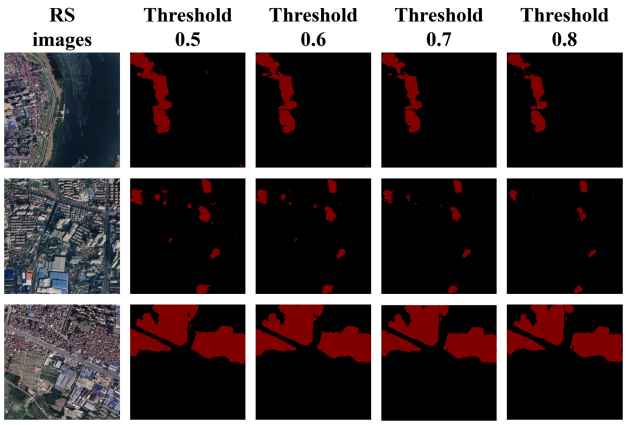



Fig. 8. Illustrations of some UIS semantic segmentation results of different confidence thresholds.

the LabelMe tool (https://github.com/wkentaro/labelme). Considering that the boundaries of UIS areas are usually not very clear, we combined high-resolution remote sensing images and auxiliary geospatial data (AOI, OSM land use surfaces, road networks, and so on) to determine the UIS boundaries when producing our semantic segmentation dataset. Finally, we obtained a UIS semantic segmentation dataset that contains 731 samples of VHR remote sensing images and UIS labels. Some of the UIS semantic segmentation samples are shown in Fig. 5. The crowdsourced weak samples used as Not UIS areas in the uncertainty-aware semisupervised learning strategy are areas of interest (AOIs) from Baidu Maps (https://map.baidu.com/) and OSM.

### C. Experimental Settings

To assess the performance of the proposed method, UNet [55], FCN [48], DeeplabV3 [52], FPN [49], FaPN [46],

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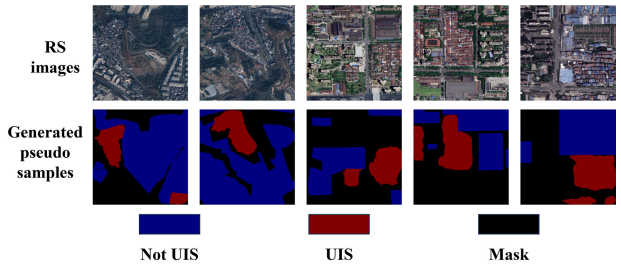



Fig. 9. Samples of the generated pseudosamples from DGUSL.

UperNet [50], SFNet [51], Segformer [43], and two semisupervised learning model, Fixmatch [53] and Unimatch [54], were adopted as comparison models in the experiments. During training, we used the Adam optimization method [56] and set the learning rate to 0.0003. The batch size was set to 16, and all models were trained with a total of 100 epochs. All implementations were based on Python (Version Python == 3.8.12). The DL framework used was PyTorch (Version Torch == 1.13.1) and TorchVision (Version Torchvision == 0.14.1). We employed an NVIDIA RTX 3090 GPU. We evaluated the UIS mapping results of the compared models using mean intersection over union (mIoU), overall accuracy (OA), and accuracy (Acc) of the UIS category and the Intersection over Union (IoU) of the UIS category, which are broadly accepted as evaluation criteria in semantic segmentation tasks. To facilitate comparisons with other models, we randomly split the total 731 samples as follows: 5% as the training set (36) and 95% as the testing set (695).

### *D. Analysis and Results of UIS Semantic Segmentation*

Table I shows the OA, mIoU, accuracy of UIS (Acc), and IoU of UIS (IoU) of different models for the YangtzeUIS dataset. It can be observed that the proposed method had the most significant performance, with nearly 0.73%–7.75% improvement in the OA, 1.71%–15.57% improvement in the mIoU, 2.78%–22.41% improvement in the Acc, and 2.63%–22.95% improvement in the IoU. Among the compared models, the Segformer model (Segformer-B1) had the most significant performance, with an OA of 89.85%, mIoU of 74.17%, Acc of 75.99%, and IoU of 60.34%. The UperNet model had the least significant performance with an OA of 82.83%, mIoU of 60.31%, Acc of 56.36%, and IoU of 40.02%. The DeepLabV3 model had a similar performance to the Segformer model. However, the number of parameters for the Segformer model was much less than that of the Segformer model. This indicates that by using the transformer and SA mechanism, the Segformer model can efficiently utilize the model parameters to learn more representative features and obtain similar results with fewer parameters. The proposed DualFormer had the second-best performance by using adaptive align shallow features from the lightweight CNN model and deep pyramidal features from the transformer model. The semisupervised compared methods, Fixmatch and Unimatch, obtained mIoU of 72.74% and 73.94%. However, due to the uncertainty in pseudosample generation, the performance cannot be improved further. The proposed SemiUIS had the most significant performance compared to all the compared models by using DGUSL. This indicates that the uncertainty-aware semisupervised learning strategy can largely improve model performance compared with the proposed DualFormer by generating large-scale UIS datasets from unlabeled data, with nearly 0.49% improvement in the OA, 1.04% improvement in the mIoU, 1.07% improvement in the Acc, and 1.52% improvement in the IoU.

We illustrate some UIS semantic segmentation results of the comparison models in Fig. 6. It can be observed that the proposed DualFormer and SemiUIS can more accurately segment informal residential areas, with fewer misclassified pixels and better geometric details of the objects (flatter geometric edges, more regular objects, and so on). For example, the results of FPN contain many internal holes and lots of misclassified salt-and-pepper noises outside of the UIS areas. The second best with the second-best performance contains many misclassified parts. The proposed DualFormer can accurately segment informal settlements but still has a few misclassified internal holes. Only the proposed SemiUIS can accurately segment informal settlements and maintain the geometric details of the UIS areas. This demonstrates the superiority of the proposed method in UIS semantic segmentation.

We further analyzed the misclassified parts of our UIS mapping results and further summarized the causes of the errors. The illustration of four representative causes that lead to UIS misidentification is shown in Fig. 7. Specifically, the not-prominent features of some UIS areas usually result in missed detection. Besides, the high intraclass heterogeneity among different UIS areas and interclass similarity between UIS and non-UIS areas typically lead to misclassification. Finally, due to the imaging conditions of the satellite sensor (such as occlusion and shooting angles), the acquired low-quality images (e.g., blurry images and areas missing) can also lead to UIS identification errors. The different occurrence of four representative causes can also explain why the UIS mapping performance varies among cities.

### *E. Ablation Experiments of Different Confidence Thresholds in DGUSL*

An ablation study is used to determine the performance of different confidence thresholds in the proposed DGUSL. The experimental results are shown in Table II. It can be seen that the model achieved the best performance when the confidence threshold is set to 0.5, with an OA of 90.58%, an mIoU of 75.88%, an Acc of 78.77%, and an IoU of 62.97%. The model obtained the least significant performance when the confidence threshold is set to 0.8, with an OA of 90.13%, an mIoU of 73.49%, an Acc of 68.31%, and an IoU of 58.45%. We further illustrate some UIS semantic segmentation results of the models with different values of the confidence thresholds in Fig. 8. It can be observed that the proposed SemiUIS with a confidence threshold of 0.5 can accurately segment informal settlements, have fewer misclassified internal holes, and maintain the geometric details of the UIS areas. Thus, in this article, the final confidence threshold is set to 0.5 in DGUSL, and a total of 4937 pseudosamples are generated.

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Fig. 10. UIS mapping results of the cities of urban agglomerations in the middle reaches of the Yangtze River.

We display some examples of the generated pseudosamples from DGUSL in Fig. 9. In this figure, the Not UIS parts are obtained from the AOI polygons of OSM and Baidu Map; the UIS parts are obtained from the confidence map. By referring to remote sensing images, it can be seen from Fig. 9 that the generated pseudosamples can well distinguish UIS areas and Not UIS areas. Besides, adding the Not UIS part of crowdsourcing data can largely reduce the uncertainty in generating pseudolabel samples. This strategy can improve the efficiency and reliability of the proposed semisupervised DGUSL.

### *F. Ablation Experiments of Different Components*

An ablation study is used to evaluate the effectiveness of various components in the proposed SemiUIS. The experimental results are shown in Table III. The blue subscript represents the decreased value. The red subscript represents the gained value. It can be observed that model1 and model2 obtained similar results, with an OA of about 84, an mIoU of about 64%, and an IoU of about 45%. By fusing features from the SFE and the PTE, model3 obtained more significant results than model1 and model2, with nearly 0.45%–0.84% improvements in the OA, 0.84%–0.95% improvements in the mIoU, and 0.65%–1.44% improvement in the IoU. This indicates that fusing features from the SFE and the PTE can help UIS mapping. By adding AFAM, the proposed DualFormer obtained the second-best performance, with a 5.26% improvement in the OA, a 10.79% improvement in the mIoU, a 15.49% improvement in the Acc, and a 15.99% improvement in the IoU compared with model3. The increment comes from the AFAM module. This proves that the accuracy of UIS mapping can be greatly improved through adaptive feature alignment and feature fusion by using the AFAM decoder. The proposed SemiUIS had the most significant performance by using DGUSL. This indicates that the uncertainty-aware semisupervised learning strategy can largely improve model performance compared with the proposed DualFormer, with nearly 0.49% improvement in the OA, 1.04% improvement in the mIoU, 1.07% improvement in the Acc, and 1.52% improvement in the IoU.

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Fig. 11. Some detailed samples of UIS mapping results of urban agglomerations in the middle reaches of the Yangtze River.

TABLE IV
VERIFICATION OF UIS MAPPING RESULTS OF URBAN AGGLOMERATIONS IN THE MIDDLE REACHES OF THE YANGTZE RIVER

| City Id | City Name | OA (%) | mIoU (%) | Acc of UIS (%) | IoU of UIS (%) | Samples |
|---|---|---|---|---|---|---|
| 1 | Wuhan City | 91.76 | 81.01 | 86.13 | 72.56 | 57 |
| 2 | Huangshi City | 93.12 | 78.65 | 73.03 | 65.21 | 22 |
| 3 | Ezhou City | 86.49 | 72.14 | 85.60 | 61.51 | 23 |
| 4 | Huanggang City | 90.19 | 76.08 | 81.63 | 64.05 | 18 |
| 5 | Xiaogan City | 91.18 | 78.21 | 86.61 | 67.20 | 23 |
| 6 | Xianning City | 89.01 | 74.10 | 88.50 | 61.54 | 29 |
| 7 | Xiantao City | 89.95 | 79.54 | 82.61 | 72.84 | 31 |
| 8 | Qianjiang City | 89.80 | 75.21 | 83.99 | 62.75 | 18 |
| 9 | Tianmen City | 91.93 | 81.20 | 83.61 | 72.67 | 6 |
| 10 | Xiangyang City | 91.69 | 81.99 | 85.82 | 75.07 | 38 |
| 11 | Yichang City | 90.83 | 69.07 | 58.30 | 48.16 | 13 |
| 12 | Jingzhou City | 87.67 | 72.62 | 73.15 | 60.44 | 30 |
| 13 | Jingmen City | 93.65 | 70.15 | 60.26 | 47.02 | 31 |
| 14 | Changsha City | 86.24 | 65.38 | 51.95 | 46.38 | 23 |
| 15 | Zhuzhou City | 88.88 | 73.84 | 70.36 | 61.15 | 8 |
| 16 | Xiangtan City | 89.72 | 69.34 | 60.69 | 50.14 | 16 |
| 17 | Yueyang City | 90.04 | 81.27 | 87.56 | 77.87 | 12 |
| 18 | Yiyang City | 93.11 | 77.04 | 89.34 | 61.82 | 4 |
| 19 | Changde City | 75.87 | 60.85 | 63.77 | 57.57 | 11 |
| 20 | Hengyang City | 86.29 | 63.37 | 47.84 | 41.97 | 10 |
| 21 | Loudi City | 94.42 | 69.38 | 51.77 | 44.61 | 8 |
| 22 | Nanchang City | 95.43 | 82.29 | 83.57 | 69.68 | 72 |
| 23 | Jiujiang City | 90.09 | 72.40 | 92.25 | 56.14 | 26 |
| 24 | Jingdezhen City | 88.93 | 71.84 | 88.15 | 56.61 | 45 |
| 25 | Yingtan City | 91.50 | 76.45 | 67.17 | 62.82 | 12 |
| 26 | Xinyu City | 91.02 | 75.50 | 66.73 | 61.48 | 4 |
| 27 | Yichun City | 92.74 | 72.63 | 79.35 | 53.18 | 16 |
| 28 | Pingxiang City | 89.33 | 70.73 | 73.22 | 53.62 | 29 |
| 29 | Shangrao City | 90.09 | 74.73 | 80.17 | 61.21 | 25 |
| 30 | Fuzhou City | 91.09 | 73.80 | 76.65 | 57.76 | 29 |
| 31 | Ji'an City | 91.59 | 70.92 | 80.06 | 51.05 | 6 |

### G. Urban Agglomeration-Scale UIS Mapping Products and Validation

We display detailed UIS mapping results of the individual cities of urban agglomerations in the middle reaches of the Yangtze River in Figs. 10 and 11. Accuracy assessment and validation of the UIS mapping results were performed in three ways.

*1) Validation by Visual Interpretation of Labeled Remote Sensing Samples:* In the validation by visual interpretation of labeled remote sensing samples, the OA, mIoU, Acc, and IoU were calculated by manually labeled testing samples in each city. The number of samples of the cities varies from 4 to 57. Each testing sample is with a length of 465 pixels and a width of 465 pixels. The results are shown in Table IV. As can be seen from Table IV, 19 of the 31 cities have OA greater than 90%. Among them, the cities with the highest and lowest OA are Nanchang City and Changde City, respectively; 25 of the 31 cities have mIoU greater than 70%. Among them, the cities with the highest and lowest mIoU are Nanchang City and Hengyang City, respectively; 22 of the 31 cities have Acc greater than 70%. Among them, the cities with the highest and lowest Acc are Jiujiang City and Hengyang City, respectively; 18 of the 31 cities have IoU greater than 60%. Among them, the cities with the highest and lowest IoUs are Yueyang City and Hengyang City, respectively.

*2) Validation by Visual Interpretation of Labeled Street View Images:* The visual interpretation using remote sensing images only may lead to UIS misjudgment; thus, we use street view images from a human perspective for more accurate verification of UIS mapping results. When verifying with street view images, we randomly selected one street viewpoint in the 15 UIS areas and one street viewpoint in the 15 non-UIS areas in the UIS mapping results for each of the 31 prefecture-level cities and obtained the interpretation result by human experts. If the street view image interpretation result is con-

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TABLE V

VALIDATION BY STREET VIEW IMAGES FOR UIS MAPPING RESULTS OF URBAN AGGLOMERATIONS IN THE MIDDLE REACHES OF THE YANGTZE RIVER

| City Name | Acc of UIS (%) | OA (%) | City Name | Acc of UIS(%) | OA (%) | City Name | Acc of UIS (%) | OA (%) | City Name | Acc of UIS (%) | OA (%) |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Shangrao City | 73.33 | 86.67 | Jiujiang City | 73.33 | 86.67 | Xiantao City | 86.67 | 86.67 | Nanchang City | 100 | 100 |
| Ji'an City | 66.67 | 83.33 | Xianning City | 86.67 | 80.00 | Tianmen City | 93.33 | 83.33 | Loudi City | 93.33 | 93.33 |
| Xiaogan City | 100 | 93.33 | Yichang City | 100 | 96.67 | Yichun City | 46.67 | 66.67 | Yueyang City | 93.33 | 93.33 |
| Changde City | 93.33 | 96.67 | Fuzhou City | 80.00 | 86.67 | Yingtan City | 73.33 | 86.67 | Jingdezhen City | 80.00 | 86.67 |
| Zhuzhou City | 66.67 | 80.00 | Wuhan City | 80.00 | 90.00 | Xiangtan City | 93.33 | 93.33 | Qianjiang City | 100 | 96.67 |
| Yiyang City | 66.67 | 80.00 | Jingzhou City | 100 | 100 | Jingmen City | 86.67 | 93.33 | Pingxiang City | 80.00 | 76.67 |
| Hengyang City | 80.00 | 90.00 | Xiangyang City | 80.00 | 83.33 | Ezhou City | 80.00 | 90.00 | Changsha City | 80.00 | 90.00 |
| Yingtan City | 73.33 | 86.67 | Huanggang City | 100 | 90.00 | Huangshi City | 100 | 100 | Total | 83.87 | 88.39 |

TABLE VI

VALIDATION BY KNOWLEDGE FOR UIS MAPPING RESULTS OF URBAN AGGLOMERATIONS IN THE MIDDLE REACHES OF THE YANGTZE RIVER

| City Name | GKP (%) | LKP (%) | City Name | GKP (%) | LKP (%) | City Name | GKP (%) | LKP (%) | City Name | GKP (%) | LKP (%) |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Shangrao City | 94.71 | 92.63 | Jiujiang City | 88.66 | 76.06 | Xiantao City | 99.31 | 98.99 | Nanchang City | 87.79 | 76.95 |
| Ji'an City | 80.92 | 80.05 | Xianning City | 92.93 | 88.95 | Tianmen City | 95.14 | 93.83 | Loudi City | 96.24 | 95.06 |
| Xiaogan City | 98.89 | 96.32 | Yichang City | 81.03 | 78.07 | Yichun City | 83.11 | 85.92 | Yueyang City | 98.18 | 96.40 |
| Changde City | 95.09 | 91.06 | Fuzhou City | 95.35 | 92.57 | Yingtan City | 85.35 | 77.66 | Jingdezhen City | 96.49 | 92.29 |
| Zhuzhou City | 79.43 | 82.33 | Wuhan City | 77.96 | 69.17 | Xiangtan City | 85.68 | 80.14 | Qianjiang City | 98.26 | 96.91 |
| Yiyang City | 91.29 | 92.80 | Jingzhou City | 97.44 | 95.82 | Jingmen City | 86.80 | 86.06 | Pingxiang City | 93.53 | 94.49 |
| Hengyang City | 95.80 | 94.53 | Xiangyang City | 94.67 | 88.74 | Ezhou City | 96.65 | 92.58 | Changsha City | 70.78 | 67.20 |
| Yingtan City | 94.57 | 90.55 | Huanggang City | 99.67 | 97.93 | Huangshi City | 96.80 | 92.22 | Total | 90.33 | 85.01 |

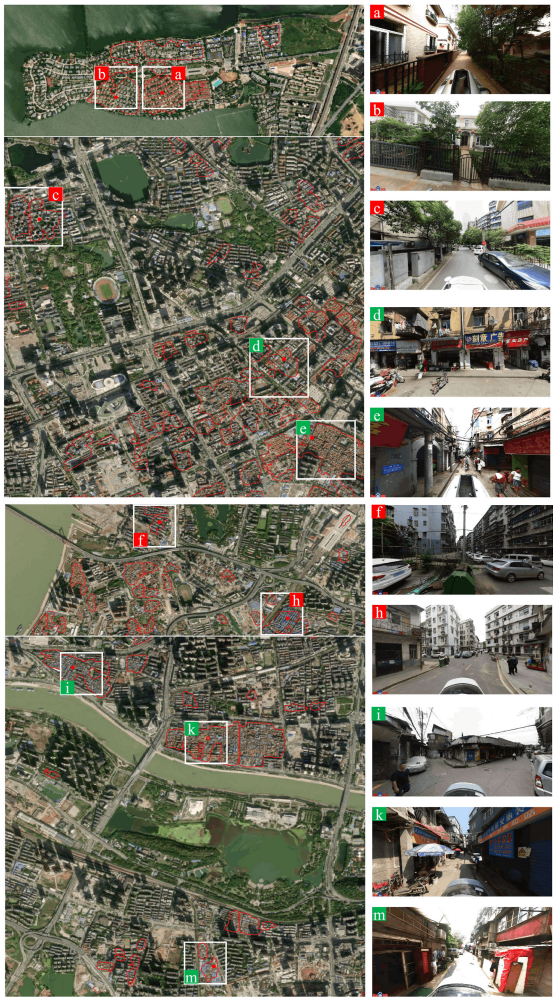



Fig. 12. Some validation examples of street view images. Green samples are correctly identified, and red samples are wrongly identified.

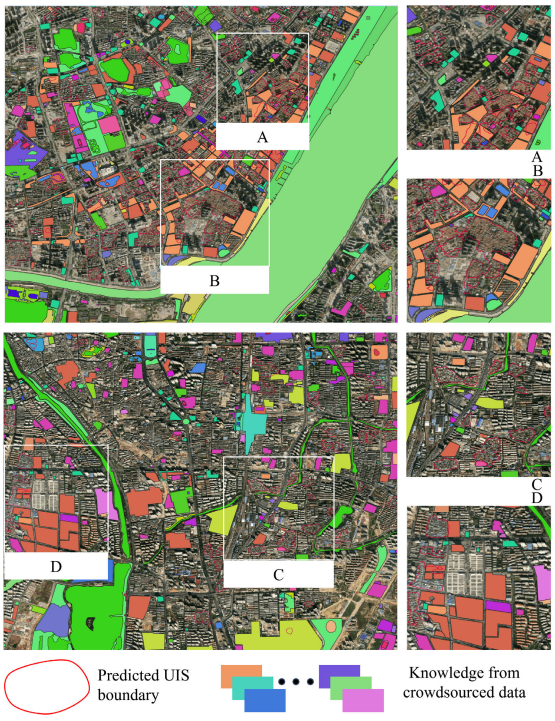



Fig. 13. Some UIS mapping results and the background knowledge from crowdsourced data.

sistent with the UIS mapping result, the UIS mapping result is considered correct; otherwise, it is considered incorrect. The results are shown in Table V. As can be seen from Table V, 16 of the 31 cities have OA greater than 90%, and 23 of the 31 cities have accuracy of UIS (Acc) greater than 80%. The mean OA and Acc of the total 31 cities are 88.39% and 83.87%. Some validation examples of street view images can be seen in Fig. 12. It can be seen that the method misclassifies low-rise villa areas as UIS areas. This is because, from the perspective of remote sensing images, these two types of objects have high feature similarity. This similarity is reflected in the building density, color, and layout patterns

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of the two types of objects, making it difficult to distinguish them from remote sensing images. It reflects the necessity of using street-view images to verify the UIS mapping results.

*3) Validation by Knowledge From Crowdsourcing Data:* As a supplement to the validation by visual interpretation of labeled samples, the GKP and LKP of the UIS mapping results are shown in Table VI. As can be seen from Table VI, 19 of the 31 cities have GKP and LKP greater than 90%, and 27 of the 31 cities have GKP and LKP greater than 80%. The total GKP and LKP are 90.33% and 85.01%, respectively. In summary, the results of validation by knowledge from crowdsourcing data and validation by visual interpretation of labeled samples jointly show that the UIS mapping results of this article have good accuracy. Some UIS mapping results with background knowledge are shown in Fig. 13.

## IV. Conclusion

In this article, a sample uncertainty-aware semisupervised learning method guided by background knowledge from crowdsourcing geospatial data, namely, SemiUIS, was proposed for refined agglomeration-scale UIS mapping. Specifically, the DualFormer model is proposed to address high spatial heterogeneity and regional mixing issues by optimizing the geometry and semantic information of UIS. Besides, a semisupervised learning strategy and an automatic sample selection method called dataset generation using uncertainty-aware semisupervised learning (DGUSL) method are proposed to select pseudosemantic segmentation samples and generate a dataset from large amounts of unlabeled data automatically. Background knowledge from crowdsourcing data is utilized to reduce the uncertainty of pseudosamples to improve the quality of the constructed dataset.

To assess the proposed model's performance, we compared it with several semantic segmentation models. Experiments were conducted in the UAMRYR, China. Visual interpretation of labeled samples, street view images, and two novel knowledge validation indicators, GKP and LKP, are proposed to verify the UIS mapping results. The proposed SemiUIS reached an OA of 90.58%, an mIoU of 75.88%, an Acc of 78.77%, an IoU of 62.97%, a GKP of 90.33%, and an LKP of 85.01%, respectively, using only 5% labeled training samples. Our method demonstrates the potential and value of DL technology and semisupervised learning with background knowledge from third-party large-scale crowdsourcing datasets in UISs mapping with few labeled samples.

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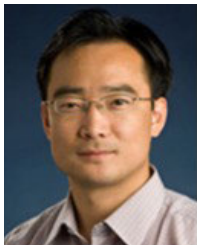
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